Image Enhancement in the Frequency Domain

Subtopic 3.2: Properties of Fourier Transform

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Outline - Subtopic 3.2: Properties of Fourier Transform

Importance of Fourier Transform Properties

- **Why Study Properties?**
  - Simplify complex signal processing tasks
  - Provide physical interpretation of operations
  - Guide efficient implementation strategies
  - Enable better understanding of image behavior
- **Key Properties to Explore:**
  - Logarithmic transformation
  - Separability
  - Translation (Shift)
  - Periodicity
  - Symmetry
- These properties form the foundation for practical frequency domain applications

Logarithmic Property: Overview

- **Problem:** The magnitude spectrum often has very large dynamic range
  - DC component (low frequency) can be orders of magnitude larger than high frequencies
  - Difficult to visualize in standard plots
- **Solution:** Apply logarithmic transformation to magnitude spectrum

$$D(u, v) = \log(1 + |F(u, v)|) \tag{1}$$

- The "+1" term prevents taking log of zero

Logarithmic Transformation: Details

Mathematical and Practical Aspects

- **General Form:**

$$g(x,y) = c\log(1 + |f(x,y)|) \tag{2}$$

  where c is a positive scaling constant

- **Properties of Logarithmic Scaling:**
  - Compresses large values while preserving small details
  - Improves visibility of low-energy frequency components
  - Makes it easier to distinguish subtle variations

- **Common Applications:**
  - Magnitude spectrum visualization
  - Power spectrum display
  - Enhanced contrast in frequency domain

Example: Logarithmic Transformation

Effect on Magnitude Spectrum

Without Log Transformation:

- Very bright DC component
- Other components barely visible
- Poor visualization
- Limited information display

With Log Transformation:

- All frequencies visible
- Better detail preservation
- Enhanced visualization
- More interpretable display

$$|F(u,v)|_{\text{original}} = 10^6 \Rightarrow |F(u,v)|_{\text{log}} \approx 6.3 \qquad (3)$$

Separability Property: Definition

- **Core Concept:** A 2-D Fourier transform can be separated into two sequential 1-D transforms
- **Mathematical Expression:**

$$F(u, v) = \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} f(x, y) e^{-j2\pi(ux/M + vy/N)} \tag{4}$$

Can be rewritten as:

$$F(u, v) = \sum_{x=0}^{M-1} e^{-j2\pi ux/M} \left[\sum_{y=0}^{N-1} f(x, y) e^{-j2\pi vy/N} \right] \tag{5}$$

Separability: Computational Advantage

- **Step-by-step Process:**
  1. Apply 1-D DFT to each **row** of the image
  2. Apply 1-D DFT to each **column** of the result
- **Computational Complexity Reduction:**
  - Direct 2-D DFT: $\mathcal{O}(M^2N^2)$ operations
  - Using Separability: $\mathcal{O}(MN(M+N))$ operations
  - With 2-D FFT: $\mathcal{O}(MN\log(MN))$ operations
- **Example:** For a 512×512 image
  - Direct: 68 billion operations
  - Separable: 5 million operations
  - With FFT: 4.7 million operations

Separability: Implementation

- **Algorithm Steps:**
  1. For each row x of image $f(x, y)$:

$$F_{\text{temp}}(u, y) = \sum_{x=0}^{M-1} f(x, y) e^{-j2\pi ux/M} \tag{6}$$

  2. For each column u of intermediate result:

$$F(u, v) = \sum_{y=0}^{N-1} F_{\text{temp}}(u, y) e^{-j2\pi vy/N} \tag{7}$$

- **Advantages:**
  - Dramatically reduces computation time
  - Standard in practical FFT implementations
  - Allows use of highly optimized 1-D FFT routines

Translation Property: Spatial Domain

- **Definition:** A shift in spatial domain causes a phase shift in frequency domain

- **Mathematical Expression:**

$$\text{If } f(x, y) \xrightarrow{\mathcal{F}} F(u, v) \tag{8}$$

Then a spatial shift gives:

$$f(x - x_0, y - y_0) \xrightarrow{\mathcal{F}} e^{-j2\pi(ux_0/M + vy_0/N)} F(u, v) \tag{9}$$

- **Key Observation:**
  - Magnitude spectrum remains **unchanged**: $|F(u, v)|$
  - Only phase spectrum changes by factor $e^{-j2\pi(ux_0/M+vy_0/N)}$

Translation Property: Frequency Domain

- **Inverse Property:** A shift in frequency domain causes modulation in spatial domain
- **Mathematical Expression:**

$$e^{j2\pi(u_0x/M+v_0y/N)}f(x,y) \xrightarrow{\mathcal{F}} F(u-u_0, v-v_0) \tag{10}$$

- **Practical Implications:**
  - Multiplying image by complex exponential shifts its spectrum
  - Used in frequency filtering and modulation
  - Foundation for heterodyning and demodulation

Translation Property: Practical Applications

- **Application 1: Frequency Shifting**
  - Move filter to arbitrary position in frequency domain
  - Useful for band-pass filtering
- **Application 2: Image Registration**
  - Phase correlation for image alignment
  - Detect shift between two images
- **Application 3: Motion Detection**
  - Track phase changes between frames
  - Identify object movement in sequences
- **Key Insight:** Spatial shifts don't change frequency content, only phase relationship

Periodicity Property: Definition

- **Core Concept:** Both discrete time and frequency sequences are periodic

- **Periodicity in Frequency Domain:**

$$F(u+M,v) = F(u,v) \quad \text{and} \quad F(u,v+N) = F(u,v) \tag{11}$$

- **Periodicity in Spatial Domain:**

$$f(x+M,y) = f(x,y) \quad \text{and} \quad f(x,y+N) = f(x,y) \tag{12}$$

- The DFT treats both spatial and frequency domains as **periodic with period** $M \times N$

Understanding Periodicity

Visualization and Interpretation

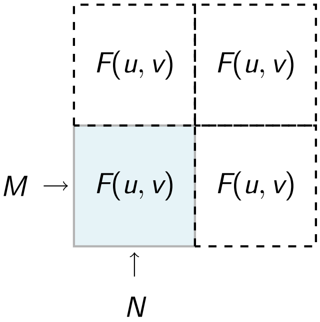
- **Conceptually:** Both domains extend infinitely as repetitions

$$f(x, y) = f(x + kM, y + lN) \quad \forall k, l \in \mathbb{Z} \tag{13}$$

- **Frequency Domain View:**
  - Frequency spectrum repeats every M samples in u-direction
  - Frequency spectrum repeats every N samples in v-direction
- **Practical Consequence:**
  - Only need to compute/display one period
  - Usually show $u \in [0, M - 1]$ and $v \in [0, N - 1]$
  - Can visualize as tiled copies of fundamental period

Periodicity Visualization

Periodic Extension of 2-D DFT



All tiles are identical copies of the fundamental period

Implications of Periodicity: Key Consequences

- **Implication 1: Aliasing**
  - If signal contains frequencies beyond Nyquist rate
  - High frequencies wrap around and corrupt low frequencies
  - Requires band-limiting and proper sampling
- **Implication 2: Circular Convolution**
  - Multiplication in frequency domain = Circular convolution
  - Must pad sequences appropriately to avoid wrap-around
- **Implication 3: Zero Padding**
  - Necessary for linear convolution using FFT
  - Prevents circular wrap-around effects

Aliasing: Mathematical Perspective

- **Nyquist Criterion:** Maximum frequency that can be represented is $f_{\text{max}} = \frac{f_s}{2}$
  - f_s = sampling frequency
  - For images: depends on pixel spacing
- **Aliasing Effect:**

$$f_{\text{alias}} = |f - kf_s| \quad \text{where } k \text{ is an integer} \tag{14}$$

- **Prevention Strategy:**
  - Low-pass filter before downsampling
  - Ensure no frequency components exceed Nyquist rate
  - Use anti-aliasing filters

Circular vs. Linear Convolution

- **Linear Convolution:** Standard convolution without wrapping

$$(f * g)(n) = \sum_{m=0}^{\infty} f(m)g(n-m) \tag{15}$$

- **Circular Convolution:** Due to periodicity in DFT

$$(f \circledast g)(n) = \sum_{m=0}^{N-1} f(m)g((n-m) \bmod N) \tag{16}$$

- **Key Difference:**
  - Circular convolution wraps around at boundaries
  - Can cause artifacts at image edges
- **Solution:** Zero Padding
  - Pad both sequences with zeros to length $M + N - 1$
  - Converts circular to linear convolution

Zero Padding Strategy

- **Why Zero Padding?**
  - Eliminates wrap-around effects
  - Enables linear convolution via FFT
  - Prevents border artifacts
- **How Much Padding?**
  - Image size: $M \times N$
  - Filter size: $P \times Q$
  - Pad to size: $(M + P - 1) \times (N + Q - 1)$
- **Practical Consideration:**
  - FFT most efficient when size is power of 2
  - May pad to nearest 2^k
  - Small overhead for efficiency gain

Symmetry Property: Hermitian Symmetry

- **Fundamental Property:** For real-valued images, FT exhibits Hermitian symmetry

$$F(-u,-v) = F^*(u,v) \tag{17}$$

where F^* denotes complex conjugate

- **Consequences:**

$$|F(-u,-v)| = |F(u,v)| \quad \text{(magnitude symmetric)} \tag{18}$$
$$\angle F(-u,-v) = -\angle F(u,v) \quad \text{(phase antisymmetric)} \tag{19}$$

- **Implications:**
  - Magnitude spectrum is **even function**
  - Phase spectrum is **odd function**

Symmetry: Even and Odd Functions

- **Even Symmetry (about origin):**

$$g(-u, -v) = g(u, v) \tag{20}$$

  Example: $|F(u, v)|$ (magnitude spectrum)

- **Odd Symmetry (about origin):**

$$g(-u, -v) = -g(u, v) \tag{21}$$

  Example: $\angle F(u, v)$ (phase spectrum)

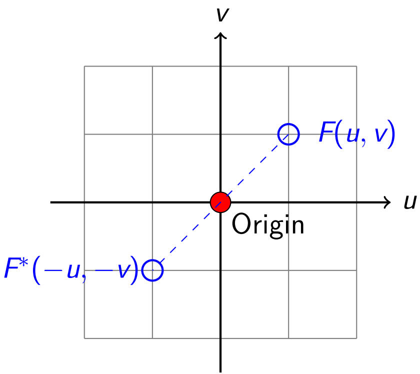
- **Practical Advantage:**
  - Only need to compute half of the spectrum
  - Other half can be obtained by symmetry
  - Reduces computation and storage

Symmetry: Computational Implications

- **Storage Reduction:**
  - For $M \times N$ real image
  - Complete spectrum needs $M \times N$ complex values
  - Using symmetry: need only half (approximately $M \times N/2$)
- **Computation Reduction:**
  - FFT algorithms exploit symmetry
  - Real FFT variants leverage Hermitian symmetry
  - Approximately $2\times$ speed improvement possible
- **Visualization:**
  - Often display only positive frequencies
  - Use quadrant shifting for centered view
  - Magnitude spectrum shows 4-fold symmetry

Symmetry in 2-D Frequency Domain

Hermitian Symmetry Visualization



Every point (u, v) in spectrum is symmetric conjugate of point $(-u, -v)$

Summary: Fourier Transform Properties

Property	Type	Key Equation/Effect
Logarithmic	Scaling	$D(u, v) = \log(1 + \|F(u, v)\|)$
Separability	Computational	2-D DFT = Two 1-D DFTs
Translation	Spatial-Freq	Phase shift by $e^{-j2\pi(ux_0/M+vy_0/N)}$
Periodicity	Both domains	$F(u + M, v) = F(u, v)$
Aliasing	Periodicity effect	Wrap-around if $f > f_{Nyquist}$
Zero Padding	Periodicity effect	Prevents circular convolution
Symmetry	Real signals	$F(-u, -v) = F^*(u, v)$

Applications of Fourier Properties

- **Frequency Domain Filtering:**
  - Use separability for efficient implementation
  - Apply periodicity-aware padding
  - Exploit symmetry to reduce computation
- **Image Enhancement:**
  - Log scaling for better visualization
  - Translation property for adaptive filtering
  - Symmetry for optimized algorithms
- **Edge Effects Management:**
  - Zero padding to prevent circular artifacts
  - Understanding periodicity implications
  - Proper boundary handling

Key Takeaways - Fourier Transform Properties

- **Logarithmic Property:** Enables better visualization of magnitude spectrum
- **Separability:** Reduces 2-D DFT to efficient sequential 1-D transforms
- **Translation:** Relates spatial shifts to phase changes in frequency domain
- **Periodicity:** Both spatial and frequency domains are periodic
- **Implications:** Aliasing, circular convolution, and zero padding requirements
- **Symmetry:** Hermitian property for real-valued images reduces computation

These properties form the foundation for efficient and practical frequency domain processing